

Sample Audio System and Design

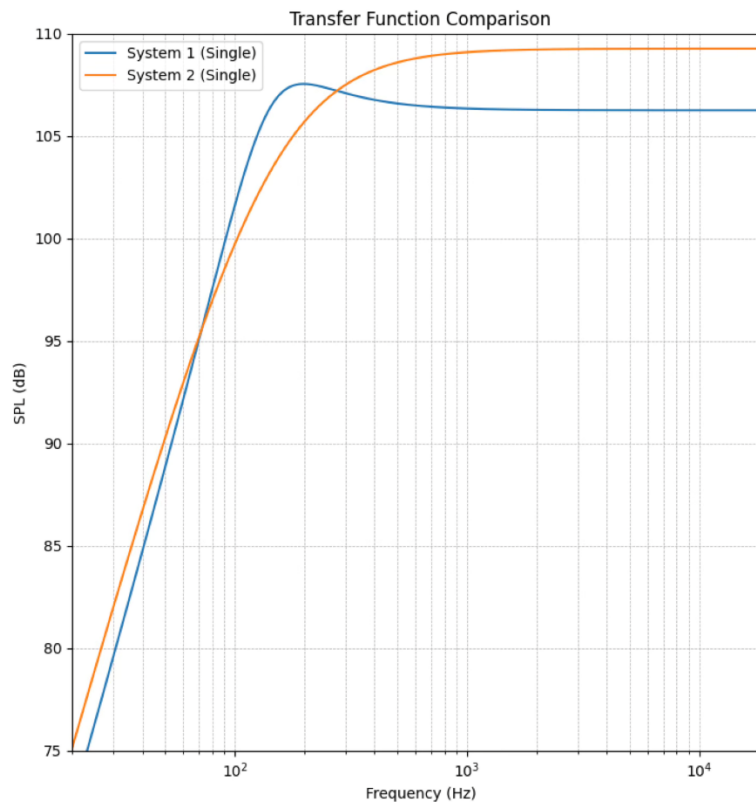
Defining the vehicle

I am tasked with working on an upcoming 2029 vehicle. The OEM client desires this model to play at a minimum of 3dB louder from 300 Hz and up. The client has expressed that they are willing to invest in the development of a new high-efficiency speaker. Naturally, raising the efficiency of a speaker by 3dB usually comes with some trade-offs, which will be investigated in this report. Other design decisions will be discussed throughout this report as well, from protecting the amp, to methods for tuning, to reliability testing.

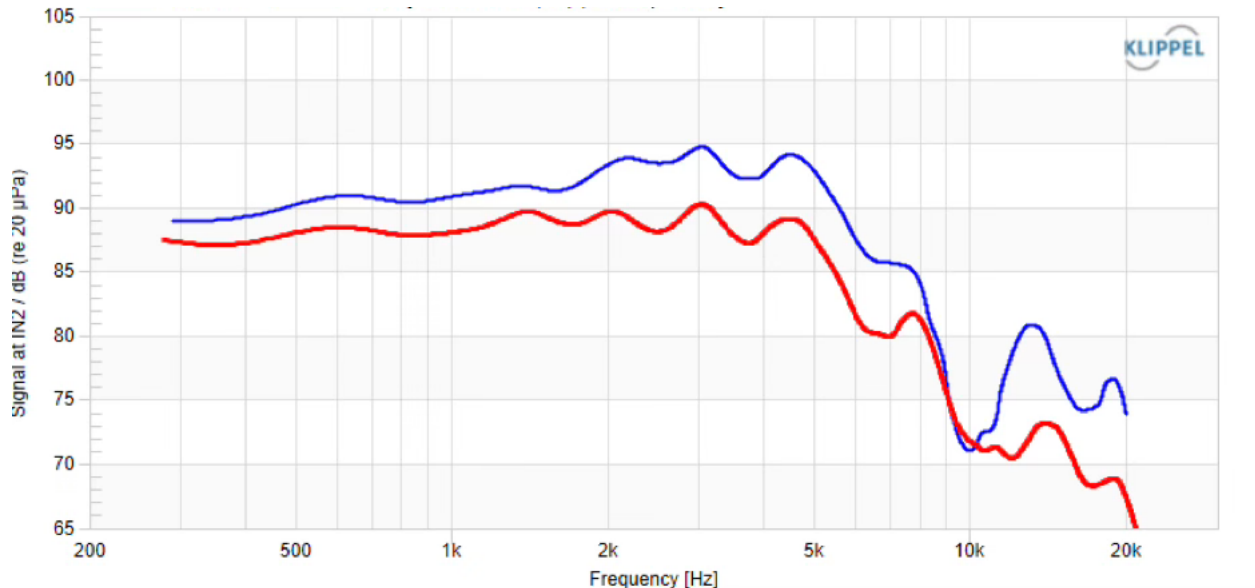
Procedure

Our loudspeaker team has come up with a few models of this new high-efficiency speaker; to do this, they used tools like FineMotor and VituixCAD.

- Given that my machine is a MAC, and there is no VituixCAD for MAC, I have built a Python alternative to VituixCAD that will allow me to see the simulated driver response. The code to this tool can be found [here](#).
- Here is the predicted response given the changes that the loudspeaker team proposed, where System 1 is MY28 and System 2 is MY29:



This simulation proved that a prototype of the 5.25 would be worth making, as long as our client could accept that there would be an attenuated response below 280 Hz or so. Our OEM client was okay with the tradeoff. The prototypes were made, and using Klippel, we measured this response:

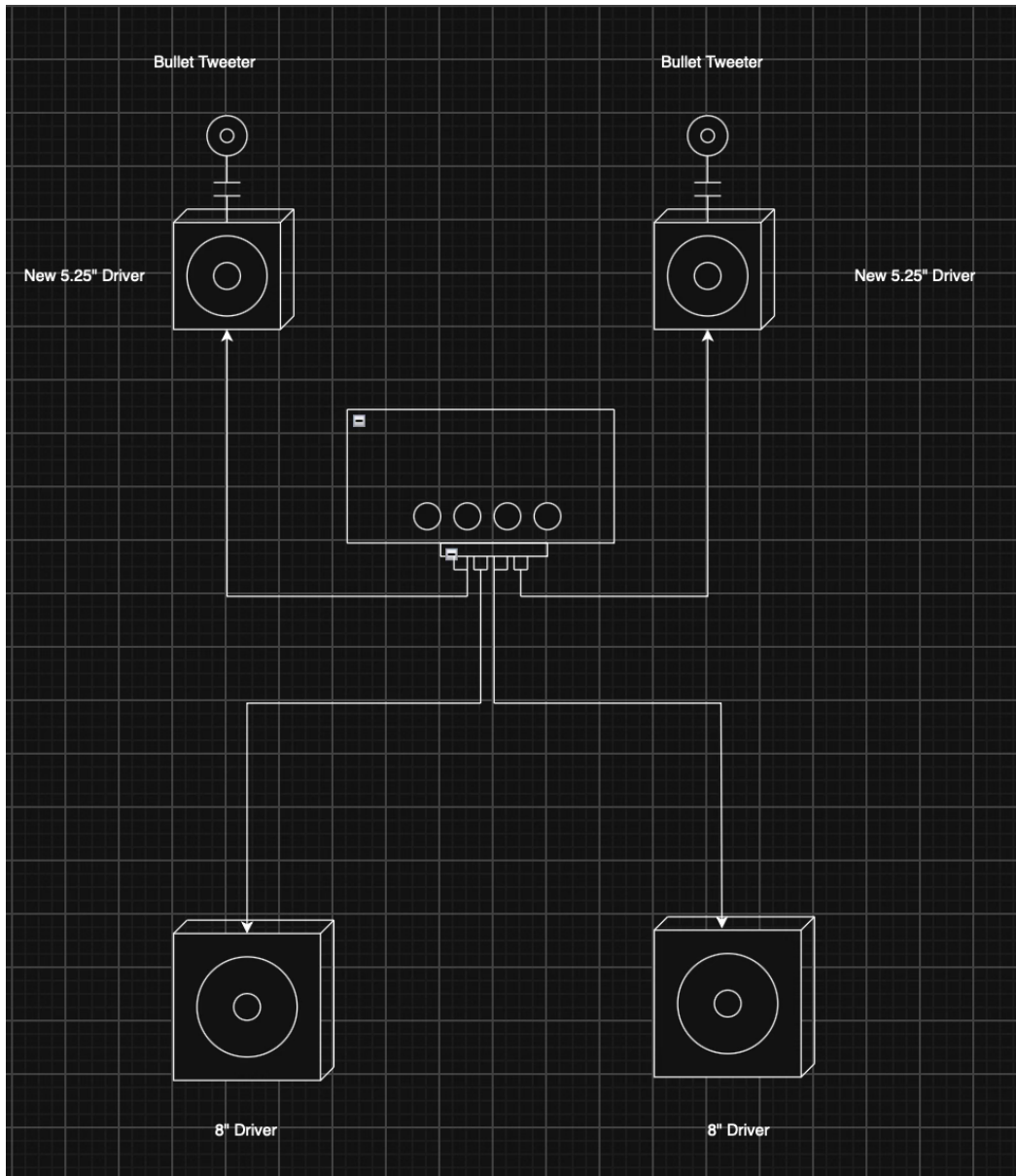


Where red is the MY28 and blue is the MY29, the response is a very much desired response for the OEM, and so a proof-of-concept vehicle will be further developed, one that they could listen to.

Next, the client expressed that their customers have complained that when driving at higher speeds, it is difficult to hear high frequencies. We proposed that we should use some bullet (a.k.a. horn) tweeters as a solution; these tweeters are highly directive and have very high sensitivity, and could provide the piercing sound that the customers are in need of.

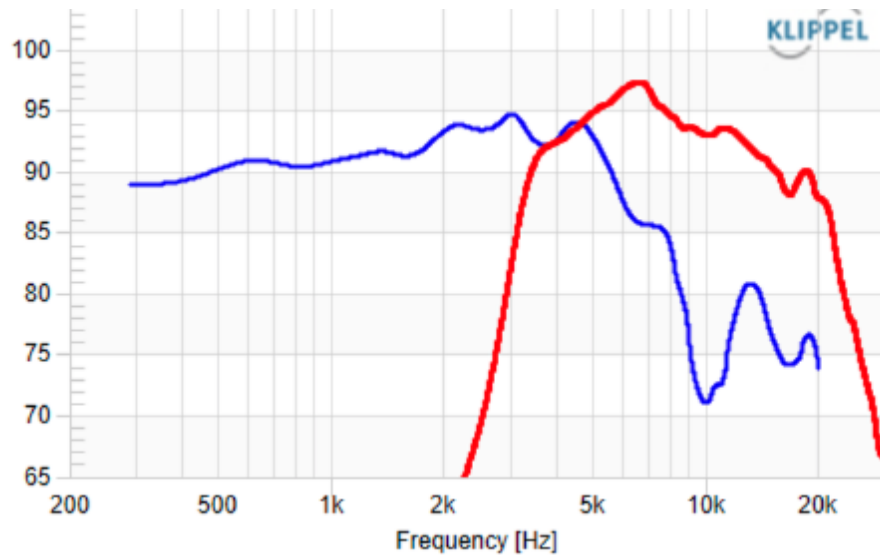
For simplicity, and to explain details easily to our OEM clients, I usually make diagrams of the proposed system and keep track of changes or issues that may exist in the system.

Some other iterations of this diagram have been made. For example, the clients expressed an idea of having a subwoofer in the rear rather than two 8" drivers, to improve the low-frequency output of the system, given that we are losing frequencies below 300 Hz due to the new 5.25" drivers. However, I pointed out to them that this specific amp is not 2-ohm rated, and bridging the two rear channels of this amp would cause the amp to see an effective 2-ohm load on the subwoofer. Given this issue, we are sticking with the previous MY28 design, with the two 8" drivers in the rear doors. Another option could have been to add a second amp, one specifically for the subwoofer, but the clients did not have the budget for the second amp, nor the new testing resources that would be required for this level of change.



The bullets are rated at 4 ohm, but given that the amp in this vehicle only has two available channels for the new 5.25" drivers, we would need to passively cross over the tweeters to protect them from playing low frequencies, which would overheat the tweeters. The loudspeaker team used VituixCAD to design a passive 2nd-order crossover set to 4.5 kHz.

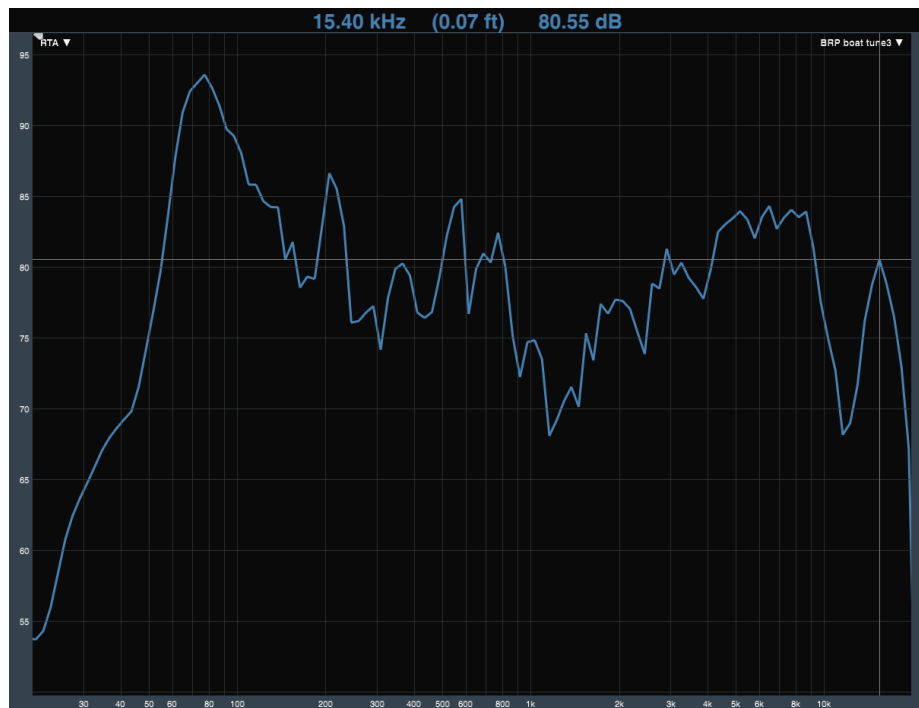
Here is a Klippel overlay of the woofer and tweeter response:



Given that the high-efficiency 5.25" drivers have a decent response all the way up to 5 kHz, it would be best to cross the tweeters over at around 4.5 to 4.8 kHz.

Proof Of Concept

It is now time to make a mock system, give it a sample tune, and demo it for the customer to verify that they indeed wish to proceed with the development of this project. After wiring the system together, I placed our 6-microphone array in the driver's seat and got this response using Rational Acoustic's Smaart v8 RTA software:



What I notice is that the cabin of the vehicle is canceling some of those middle frequencies; one can notice the extreme cut at 1 kHz. It might be possible to slightly fix this in the tune, but not much can really be done in this case. The tweeters do have a nice response, and the bass definitely can be improved. This likely will be done using the rear channels, as the 8" drivers with more surface area can produce this lower-frequency output.

Before any tuning, it's best to set the crossovers of the woofers; we do not need to do this for the tweeters, since they already have passive crossovers on them. I set an LPF for the 8" drivers at around 1.2 kHz. The high-efficiency 5.25" drivers would then have an HPF crossover at around 1 kHz, and an LPF crossover at 4.8 kHz, as we mentioned earlier. These crossovers allow the drivers to focus more of their "energy" on the frequencies they play best. Since we have the high-efficiency drivers playing from 280 Hz and up, there is no need for the 8" drivers to be playing 2 kHz, for example, allowing the 8" drivers to focus on playing the lower frequencies that the 5.25" drivers cannot. Likewise, the same logic applies to the 5.25" drivers: there is no need for them to play 5 kHz and up when the horn tweeters can already do this so well; it's best to focus their energy toward those middle frequencies they play best.

I will now begin to state the problems I see in the frequency response, and the various filters, gains, and Q values I applied to tackle them. My approach is always to cut first, starting with the lower-frequency issues, as fixing lower-frequency problems can have harmonics that affect the high-frequency response, thus eliminating the need to work twice or thrice as much, saving computational efficiency on the DSP, and generally leading to a better-sounding system too.

The sharp peak at 200 Hz was my first point of attention. Using my Q calculator Excel sheet, which can be found [here](#), I can plug in various frequency values, such as F1 (the lower frequency of the peak or null), F2 (the higher frequency of the peak or null), and Fc (the center frequency of the peak or null). The general way of finding F1 or F2 is to look at Fc and find frequencies that are 3 dB down on either side. This is not always possible, and it may not be what you want, which is why I have three calculation methods in my sheet. Option 1 allows the user to input F1 and F2, and the sheet will calculate Fc and Q. Option 2 allows the user to input F1 and Fc, and the sheet will calculate F2 and Q. Option 3 allows the user to input F2 and Fc, and the sheet will calculate F1 and Q.

The F1 and F2 of this peak are located at 183 Hz and 244 Hz respectively, meaning the Fc is at 211 Hz and the Q is 3.46. Looking at the graph, this peak appears to be about 6 dB above the average of the rest of the curve, so the gain was set to -6 dB.

The next peak was at 580 Hz. The F1 was about 485 Hz, the F2 was calculated to be at around 700 Hz, and the Q was calculated to be 2.78. This peak was about 3 dB above the average of the trace, so the gain was set to -3 dB.

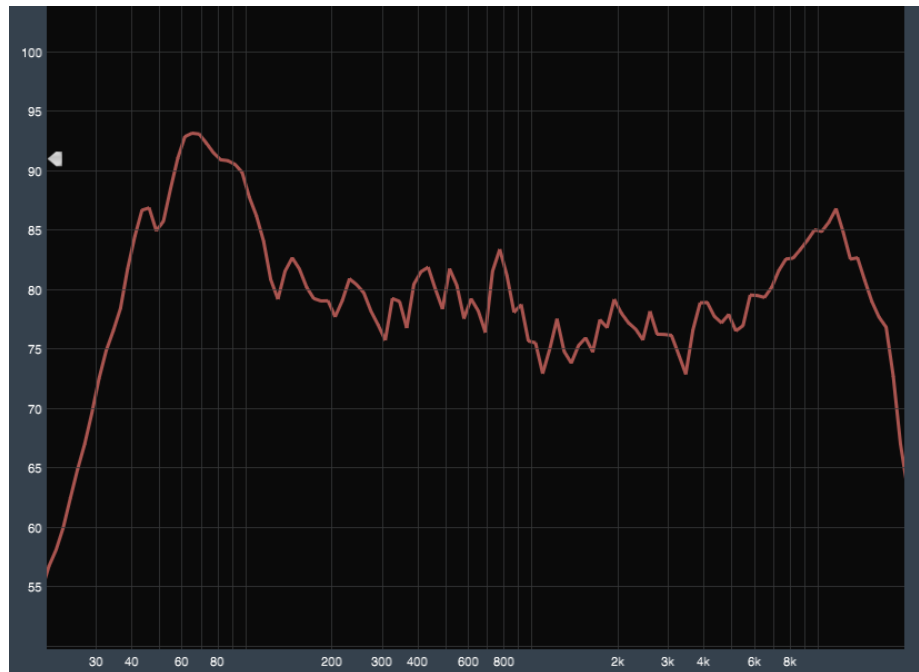
The next point was the band of frequencies between 2.6 kHz and 3.8 kHz; its Fc was 3170 Hz and its Q was 2.49. The whole band needed to be lowered by about 5 dB.

Lastly, there is another band of frequencies that is a bit too high from 4 kHz and up. A high shelf filter with -7 dB gain was placed to flatten out this curve.

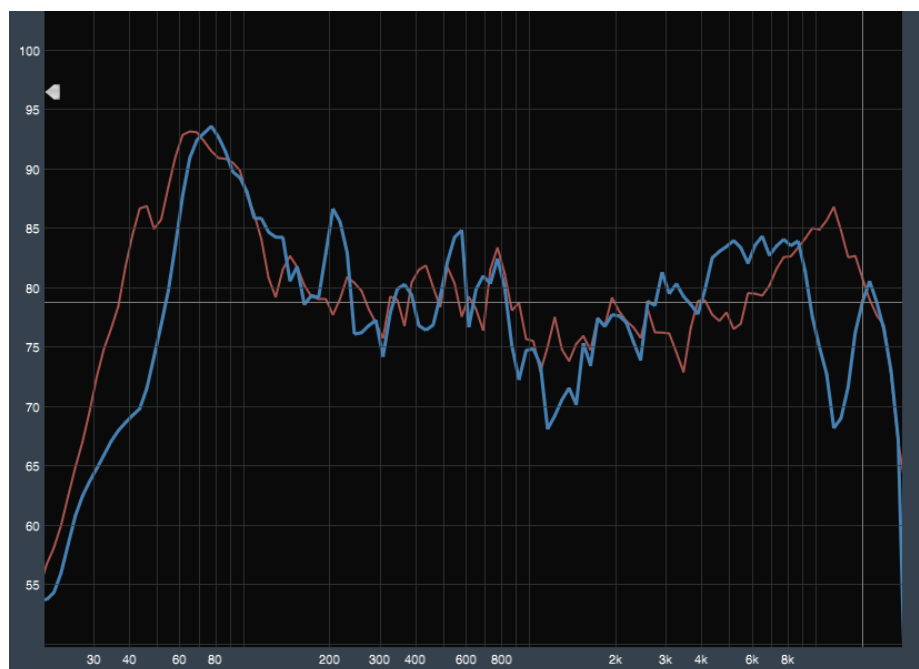
Once all the cuts were made, I tried to gain back the 1 kHz null that we talked about earlier. The F1 of this null is at around 870 Hz and the F2 is at 1650 Hz, meaning the Fc is at about 1200 Hz and the Q is 1.53. I added a gain of +5 dB, but as I mentioned earlier, this was clearly a null; the actual track only budged about 2 to 3 dB with this change.

After listening to the output, I was reminded that one of the asks from the OEM client was for the high-frequency response to cut through the wind noise, so I added a high shelf filter at around 6 kHz and up to get the piercing sound to cut through the wind and engine noise.

The final curve looked like this:



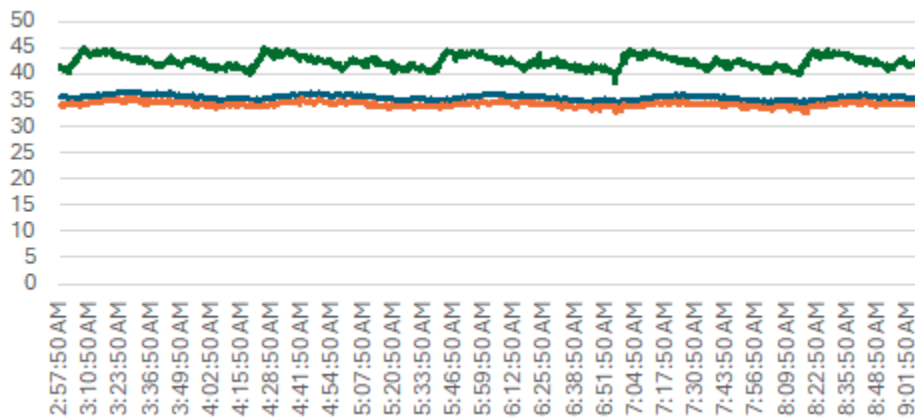
Here is a before and after comparison:



Reliability Testing:

The last step in this whole process, before I could send out this tune for OEM client approval, is to stress test it. I take the entire system and the tune I gave it, then play a list of very loud, bass-heavy songs at max volume and run the whole system for 36 hours. Thermocouples are added to the yoke of the drivers and connected to a data logger that records the thermal data of the entire test. This also allows me to see at what point any drivers in the system failed, as they cool down shortly after failing.

The stress test was run for 36 hours with no problems. The thermal data of the drivers is shown here:



Time vs. Temperature (Celsius) graph

Where the green trace is one of the 8" drivers, and the blue and orange are the 5.25" high-efficiency drivers.

All the drivers passed the test, and the tune and proof-of-concept system can be sent out to our OEM client for review and feedback.